Apollo really about?

The scope and audacity of John Kennedy's May 25, 1961, message to a joint session of Congress on "Urgent National Needs"—the speech that launched the *Apollo* program—dazzled me. We would use rockets not yet designed and alloys not yet conceived, navigation and docking schemes not yet devised, in order to send a man to an unknown world—a world not yet explored, not even in a preliminary way, not even by robots—and we would bring him safely back, and we would do it before the decade was over. This confident pronouncement was made before any American had even achieved Earth orbit.

As a newly minted Ph.D., I actually thought all this had something centrally to do with science. But the President did not talk about discovering the origin of the Moon, or even about bringing samples of it back for study. All he seemed to be interested in was sending someone there and bringing him home. It was a kind *of gesture*. Kennedy's science advisor, Jerome Wiesner, later told me he had made a deal with the President: If Kennedy would not claim that *Apollo* was about science, then he, Wiesner, would support it. So if not science, what?

The *Apollo* program is really about politics, others told me. This sounded more promising. Nonaligned nations would be tempted to drift toward the Soviet Union if it was ahead in space exploration, if the United States showed insufficient "national vigor." I didn't follow. Here was the United States, ahead of the Soviet Union in virtually every area of technology—the world's economic, military, and, on occasion, even moral leader—and Indonesia would go Communist because Yuri Gagarin beat John Glenn to Earth orbit? What's so special about space technology? Suddenly I understood.

Sending people to orbit the Earth or robots to orbit the Sun requires rockets—big, reliable, powerful rockets. Those same rockets can be used for nuclear war. The same technology that transports a man to the Moon can carry nuclear warheads halfway around the world. The same technology that puts an astronomer and a telescope in Earth orbit can also put up a laser "battle station." Even back then, there was fanciful talk in military circles, East and West, about